A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 - Generated practice paper Set 06

## Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance | Marks |
|---|---|---|
| **01.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere)<br>**1** They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone<br>**1** The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **ozone layer** → the ozone in the stratosphere that absorbs UV<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.<br>• answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated | 3 |



Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **01.2** | **Any 2 of 5**<br>1 mark each for any 2 of the 5 points below, in any order, maximum 2.<br>**1** Provides carbon, oxygen, hydrogen and nitrogen (as $N_2$, $O_2$, $CO_2$, $H_2O$) for biological molecules<br>**1** Delays the escape of infrared energy, keeping the surface warm<br>**1** Distributes heat and water vapour by winds / ocean currents<br>**1** Provides atmospheric pressure so liquid water can exist<br>**1** Deflects damaging radiation of the solar wind in the upper atmosphere<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **infrared** → heat radiation / long-wave radiation re-emitted by the Earth<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| **01.3** | 1 mark for the point below.<br>**1** Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **stratified** → sampling each area or group in proportion to its size | 1 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| 01.4 | **Levels of response**<br>Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist.<br>**Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.<br>**Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.<br>**Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.<br>**0 marks**: No relevant content.<br>**Indicative content - a guide to placing the mark, not a checklist**<br>• **Troposphere:** enhanced greenhouse effect - anthropogenic $CO_2$ (fossil fuels, deforestation), methane (rice, livestock, landfill), $N_2O$, CFCs, tropospheric ozone<br>– more infrared absorbed and re-emitted<br>– aerosols and smoke increase albedo / scatter light<br>– land-use change alters surface albedo and evapotranspiration<br>• **Stratosphere:** CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / $O_2$ / $O_3$ equilibrium<br>– less UV absorbed in the stratosphere, more reaches the surface<br>– stratospheric cooling<br>• **Consequences:** changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions<br>– increased UV damage to organisms<br>• **Difficulties of prediction:** linked processes where one change alters others<br>– feedback (positive and negative) and tipping points<br>– time lags (ocean heat capacity, long residence times of CFCs and $CO_2$)<br>– natural variability (solar output, volcanoes, Milankovitch cycles)<br>– incomplete understanding of atmospheric processes<br>– limited historical data so proxies needed<br>– models with different assumptions<br>• **Evaluation:** some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain<br>– precautionary principle<br>• Justified conclusion<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• Any accurate, relevant material that is not in the indicative content. | 9 |
| | • **enhanced greenhouse effect** → extra warming caused by added greenhouse gases<br>• **anthropogenic** → human-caused / caused by human activity<br>• **CFCs** → CFCs / halogenated organic compounds releasing chlorine<br>**Do not credit**<br>• answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated<br>• 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | **Model answer - Level 3 (8 marks)** | |
| | In the troposphere human activity enhances the greenhouse effect: fossil fuels and deforestation add $CO_2$, rice, livestock and landfill add methane, fertilisers add nitrous oxide, and CFCs and tropospheric ozone are new absorbers. More outgoing infrared is absorbed and re-emitted, warming the surface, while smoke and aerosols scatter sunlight and land-use change alters albedo and evapotranspiration. In the stratosphere, CFCs, halons and HCFCs release chlorine and bromine that catalyse ozone destruction, shifting the O / $O_2$ / $O_3$ equilibrium so that less UV is absorbed there and more reaches the surface, cooling the stratosphere as well. | |
| | The consequences run through every system the atmosphere's energy drives: climate, thermohaline circulation, the hydrological cycle, sea level and species distributions, plus UV damage to organisms. Predicting them is hard because the processes are linked - one change alters others - and because feedback can amplify (albedo loss, methane release) or damp (cloud cover, photosynthesis) the effect, sometimes past tipping points. Time lags from ocean heat capacity and the long residence times of $CO_2$ and CFCs, natural variability from solar output, volcanoes and orbital cycles, sparse historical data replaced by proxies, and models built on different assumptions all add uncertainty. | |
| | Some things are nevertheless well established: the physics of infrared absorption, the chemistry of ozone loss, and the success of the Montreal Protocol. What is uncertain is the scale and timing of regional impacts, which is exactly why the precautionary principle, not certainty, should guide action. | |
| | **Weaker answer - Level 1 (3 marks)** | |
| | Humans release greenhouse gases like carbon dioxide from burning fossil fuels into the troposphere, which traps more heat and causes global warming. CFCs go into the stratosphere and destroy the ozone layer, so more UV reaches the Earth. It is hard to predict what will happen because the atmosphere is complicated and scientists do not know everything about it. There could be feedback effects that make it worse. | |
| | **What makes the difference** | |
| | • Level 1 gives one cause per layer; Level 3 lists several gases and processes and explains the mechanism (absorption and re-emission; catalytic destruction shifting an equilibrium). | |
| | • 'Difficulties of prediction' is answered in Level 3 with specific reasons (linked processes, feedback, time lags, natural variability, proxy data, models); Level 1 says 'complicated'. | |
| | • Level 3 evaluates - separating what is certain from what is not - and links this to the precautionary principle. | |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **02.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Many different organic substances may be present, each with a different potential to cause deoxygenation, so it is difficult to measure or forecast them individually<br>**1** BOD measures the effect (amount of oxygen used by microbes decomposing the organic matter) under standardised conditions (eg 5 days, 20 °C, dark)<br>**1** It gives a single measure directly related to the risk of deoxygenation<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **Biochemical Oxygen Demand** → the oxygen used by aerobic decomposers in a water sample<br>• **deoxygenation** → loss of dissolved oxygen from the water<br>**Do not credit**<br>• 'a high BOD means the water contains a lot of oxygen' - a high BOD means a lot of oxygen will be removed | 3 |
| **02.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** E. coli is a common gut bacterium so its presence confirms faecal contamination and its number indicates how serious the contamination is<br>**1** Pathogens are not always present so their absence would not prove the water is uncontaminated<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **02.3** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1** Different species have different sensitivities to pollution<br>**1** Easy to identify<br>**1** Easy to find / normally present / usually common<br>**1** Widely distributed<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **biotic** → living factors / effects of other organisms<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **02.4** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1 Advantage:** assesses the effect of pollution over a period of time (recent pollution events are detected even if the pollutant has gone)<br>– no expensive equipment needed<br>**1 Disadvantage:** identification needs skill<br>– does not identify which pollutant is present so further chemical tests are required<br>– species may be absent for other reasons<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **biotic** → living factors / effects of other organisms | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9. | |
| **03.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Readings drift so they may not be accurate<br>– calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions<br>**1** pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly<br>**1** If a meter cannot be adjusted, record the percentage error and correct the results<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **accurate** → close to the true value | 3 |
| **03.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Same height above the ground<br>**1** No obstacles to air flow<br>– impeller type facing into the wind<br>**1** Axis of rotation horizontal (impeller) or vertical (cup)<br>– same time period / averaging<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |
| **03.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling<br>**1** Evaporation cools the wet bulb more when humidity is low<br>– the temperature difference is compared with a table to find humidity<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **03.4** | **Any 2 of 3**<br>1 mark each for any 2 of the 3 points below, in any order, maximum 2.<br>**1** Test strips with reagent bands compared with a colour chart (semi-quantitative)<br>**1 Electronic colourimeter:** reagent added and colour intensity measured at a specific wavelength<br>**1** Ion-selective electrode measuring electrode potential (other ions can interfere)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1. | |
| **04.1** | **Calculation**<br>1 mark for the correct answer; working does not have to be shown.<br>**1** 4.2 / 0.7 = 6<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (times) is already printed on the answer line.<br>**Do not credit**<br>• A bare wrong answer with no working. | 1 |
| **04.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Long summer days (almost 24 hours of daylight) so the total energy received is high, even though the Sun is low<br>**1** In winter days are very short / the Sun does not rise, so light levels are extremely low<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **04.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Reflectors can only focus parallel rays of direct sunlight onto the absorber<br>**1** Clouds, smoke and dust scatter the light so it cannot be focused and intensified<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **04.4** | 1 mark for the point below.<br>**1** Very large areas of collectors are needed to harness significant amounts of energy (and the changing angle of incidence reduces it further)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| **04.5** | **Levels of response**<br>Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist.<br>**Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.<br>**Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.<br>**Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.<br>**0 marks**: No relevant content.<br>**Indicative content - a guide to placing the mark, not a checklist**<br>• **Problems:** intermittency (daily and seasonal cycles), unpredictable cloud cover, low energy density, changing angle of incidence, land requirements, water for cleaning in deserts, toxic wastes in manufacture (silicon tetrachloride, cadmium), efficiency loss at high temperature<br>• **Photothermal:** evacuated tubes reduce conduction / convection losses<br>– thermal stores (water, sand, concrete, molten salt) allow later use<br>– passive solar architecture, brise soleil, light tubes<br>– heat pumps deliver up to four times the energy used<br>• **Photovoltaic developments:** multijunction cells absorbing more wavelengths, anti-reflective / moth-eye textured surfaces (smooth cells reflect 30%), PV-thermal hybrids cooling the cell, transparent PV in windows, heliostats keeping the optimum angle, self-cleaning nanohydrophobic surfaces<br>• Concentrating solar power with molten-salt storage to 550 °C so steam turbines run when needed - overcomes intermittency<br>• **Integration:** grid-connected solar farms, rooftop space avoiding land conflicts, batteries, pumped storage, links to energy storage and demand management<br>• **Evaluation:** strong in sunny deserts / low latitudes, weaker in high latitudes in winter<br>– cost falling<br>– manufacturing impacts<br>– no single technology solves all problems<br>– justified conclusion<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• Any accurate, relevant material that is not in the indicative content. | 9 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | • **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous<br>• **pumped storage** → water pumped uphill when demand is low and released to generate at peak<br>**Do not credit**<br>• 'solar panels do not work in cold countries' - output depends on insolation, and cells are slightly more efficient when cool<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | |
| | **Model answer - Level 3 (9 marks)** | |
| | Solar power's problems are intrinsic: it is intermittent with daily and seasonal cycles, cloud makes it unpredictable and scatters light so it cannot be focused, its low energy density demands large collector areas, the angle of incidence keeps changing, cleaning in deserts uses scarce water, manufacturing PV produces toxic wastes such as silicon tetrachloride and cadmium, and PV efficiency falls as cells heat up. | |
| | New technology addresses most of these. Evacuated-tube photothermal panels cut conduction and convection losses, and thermal stores of water, sand, concrete or molten salt hold heat for later; passive solar design, brise soleil and light tubes reduce demand; heat pumps deliver up to four times the energy used to run them. Multijunction PV cells absorb more wavelengths, moth-eye anti-reflective surfaces stop the 30% of light a smooth cell reflects, PV-thermal hybrids cool the cell and use the heat, transparent PV turns windows into generators, heliostats keep panels at the optimum angle and nanohydrophobic surfaces self-clean. Concentrating solar power with molten salt stored at 550 °C runs steam turbines when electricity is wanted, directly overcoming intermittency, and grid-connected farms, rooftops that avoid land conflicts, batteries and pumped storage integrate the output. | |
| | Even so, no technology changes the fact that winter insolation at high latitudes is tiny, that CSP needs cloudless skies, or that manufacturing has impacts. Solar can dominate supply in sunny, low-latitude regions and contribute strongly elsewhere in summer, but overcoming its problems fully depends on storage and a diverse energy mix rather than on solar technology alone. | |
| | **Weaker answer - Level 1 (2 marks)** | |
| | Solar power has problems because it does not work at night or when it is cloudy, and solar panels are not very efficient. New technologies can help. Better solar panels are more efficient. Solar power can be stored in batteries for use at night. Mirrors can concentrate the sunlight to make it hotter. So new technologies can overcome most of the problems with solar power. | |
| | **What makes the difference** | |
| | • Level 1 identifies two problems and three vague technologies; Level 3 names each technology and explains which problem it solves and how. | |
| | • Level 3 acknowledges what technology cannot fix (high-latitude winters, cloud for CSP, manufacturing impacts) - the 'extent' in the question. | |
| | • Numbers (30% reflection, 550 °C, four-fold heat-pump output) show detailed knowledge applied to the argument. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1. | |
| **05.1** | **Correct answer only**<br>1 mark for the correct option. No mark if more than one box is ticked.<br>**1 A:** Its total annual discharge | 1 |
| **05.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Public supplies cannot match consumer demand / the water is often cleaner than rivers<br>**1** It also reduces urban flooding after heavy rain<br>– surplus can be used for aquifer recharge<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **aquifer** → a permeable rock that stores groundwater / underground water store | 2 |
| **05.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** The reservoir is created in the estuary so no farmland / settlements need to be flooded<br>**1** But it changes important inter-tidal habitats, obstructs shipping and can be polluted by activities anywhere in the river's catchment<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-123; AQA spec 3.2.2.5. | |
| **06.1** | **Calculation**<br>1 mark for the correct answer; working does not have to be shown.<br>**1** (120 + 95) - (2 + 45) = 168 t $ha^{-1}$<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (t $ha^{-1}$) is already printed on the answer line.<br>**Do not credit**<br>• A bare wrong answer with no working. | 1 |
| **06.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO2<br>**1** Harvest removes biomass so less dead organic matter is returned<br>– pasture roots and litter constantly add organic matter<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **06.3** | **Any 2 of 4**<br>1 mark each for any 2 of the 4 points below, in any order, maximum 2.<br>**1** Low / zero tillage<br>**1** Adding manure, compost or mulch<br>**1** Growing cover crops / leaving crop residues<br>**1** Converting to long-term grass or tree crops in rotation<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2. | |
| **07.1** | **2 marks per point**<br>2 marks for each of the 3 points below; award 1 mark where a point is only partly made.<br>**2** Aerodynamic shapes / fairings reduce friction with the air (air resistance) so less energy is needed to propel the vehicle, especially at speed<br>– hydrodynamics for ships<br>**2** Reduced mass (plastics, carbon-fibre composites, high-strength steel alloys, aluminium engine blocks, lighter batteries and magnets, rounded shapes) means less energy to accelerate and less rolling resistance, so lower fuel consumption<br>**2** Conventional friction brakes convert kinetic energy to waste heat<br>– regenerative braking uses a generator to convert it to electricity stored in a battery (or returned to the rail grid) which then propels the vehicle, so less fuel is needed to accelerate again<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 6 |
| **07.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Pneumatic tyres deform as they rotate<br>– the continual change in shape and air movement inside generates frictional heat (more if under-inflated)<br>**1** Steel wheels do not deform so rolling resistance is very low<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **07.3** | **Any 2 of 4**<br>1 mark each for any 2 of the 4 points below, in any order, maximum 2.<br>**1** More valves per cylinder to remove exhaust gases fully so they do not mix with fresh fuel<br>**1** Better temperature control (thermostatic fan)<br>**1** Electronic ignition control / regular servicing to keep spark timing correct<br>**1** Turbochargers supplying more oxygen<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-227; AQA spec 3.3.4.2.1. | |
| **08.1** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** The void fills with water naturally<br>– wetlands are uncommon / valuable habitats in most regions<br>**1** The new habitat may have greater wildlife value than the farmland lost<br>– low cost compared with filling and levelling<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **08.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1 Housing:** ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer)<br>**1 Farmland:** landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **08.3** | 1 mark for the point below.<br>**1** Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost<br>– transplanted plants often fail<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| 09.1 | **Levels of response**<br>Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist.<br>**Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.<br>**Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.<br>**Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.<br>**0 marks**: No relevant content.<br>**Indicative content - a guide to placing the mark, not a checklist**<br>• **Control of greenhouse gases:** energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture<br>• **Geoengineering:** white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences<br>• **International agreements:** Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation<br>– carbon trading and taxes<br>– comparison of per capita carbon footprints between countries<br>• **Evidence of limited success:** global CO2 concentrations still rising (about 280 ppm pre-industrial to over 410 ppm)<br>– temperatures up about 1 °C<br>– emissions in some developed countries falling but rising in developing economies<br>• **Reasons for difficulty:** fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop<br>• Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause<br>• Comparison with the Montreal Protocol's success for ozone<br>• **Judgement:** partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise<br>– tipping points make urgency greater<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 9 |
| | • Any accurate, relevant material that is not in the indicative content.<br>• **energy conservation** → using less energy, eg by insulation or behaviour change<br>• **CCS** → capturing carbon dioxide from flue gases and storing it underground<br>**Do not credit**<br>• 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do<br>• 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | **Model answer - Level 3 (8 marks)** | |
| | Strategies fall into control of greenhouse gases, geoengineering, international agreements and adaptation. Emissions are cut by energy conservation in transport, buildings and industry, by replacing fossil fuels with renewables and nuclear, by carbon capture and storage, by reducing deforestation and planting forests, and by capturing methane from landfill. Geoengineering ideas - white roofs, ocean fertilisation, solar shades - remain largely untried with unpredictable side effects. Adaptation, such as flood defences and managed retreat, reduces impacts but not the cause. | |
| | The Kyoto Protocol and Paris Agreement set targets, but they are weak or non-binding and participation is uneven; carbon trading and taxes exist but per capita footprints still differ enormously between countries. The evidence of success is limited: atmospheric $CO_2$ has risen from about 280 ppm before industrialisation to over 410 ppm and temperature by about 1 °C, and although emissions are falling in some developed countries they are rising in developing economies. | |
| | The difficulty is that fossil fuels underpin economies, so control demands lifestyle change, cost, and political agreement between nations with different priorities, while time delays and natural fluctuations mask the effect of any action. By contrast the Montreal Protocol succeeded because ozone was a substitution problem. The judgement must be partial success: the technology exists and regional reductions prove it works, but global strategies have not yet stopped the rise, and tipping points make the delay dangerous. | |
| | **Weaker answer - Level 2 (5 marks)** | |
| | Methods to reduce climate change include using renewable energy such as wind and solar instead of fossil fuels, saving energy, and planting trees to absorb carbon dioxide. International agreements like the Kyoto Protocol and the Paris Agreement have set targets for countries to reduce emissions. However carbon dioxide levels are still rising, so the methods have not been very successful so far. This is because countries rely on fossil fuels and do not always keep to the agreements. More needs to be done. | |
| | **What makes the difference** | |
| | • Level 2 covers the main strategies and reaches a conclusion, but with no quantitative evidence (280 to 410 ppm, 1 °C) and no mention of geoengineering or adaptation. | |
| | • Level 3 explains *why* success is limited (economic dependence, time lags, natural variability, political disagreement) rather than just stating that it is. | |
| | • The Montreal comparison and the tipping-point point give Level 3 the breadth and judgement the top band requires. | |
| **09.2** | 1 mark for the point below.<br>**1** The total greenhouse gas / CO2 emissions caused by an individual, organisation, product or country (often expressed per capita per year)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2. | |
| **10.1** | **Levels of response**<br>Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist.<br>**Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.<br>**Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.<br>**Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.<br>**0 marks**: No relevant content.<br>**Indicative content - a guide to placing the mark, not a checklist**<br>• **Features:** renewable / non-renewable, depletable, abundance, energy density, locational constraints, intermittency and predictability, need for conversions, applicability, ease of storage and transport, technological development, political and economic influences, environmental impacts<br>• **Fossil fuels:** high energy density, easy storage and transport, reach high temperatures, well-developed technologies and infrastructure - but finite, polluting (acid rain, smoke, $CO_2$), politically concentrated (Middle East oil)<br>• **Nuclear:** very high energy density, base load, few fuel deliveries - but complex, expensive, waste, public opposition<br>• **Renewables:** low energy density, intermittent (wind, solar unpredictable; tidal predictable), locationally constrained, cannot be stored without conversion, need storage / grid, cannot reach fossil-fuel temperatures or replace liquid vehicle fuels yet - but renewable, low pollution, falling costs<br>• **Sustainability concepts:** impacts before use (extraction, processing, equipment embodied energy) and as a consequence of use (pollution, habitat damage, depletion)<br>• **Economic and political factors:** subsidies, guaranteed prices, planning, external costs not included in price, energy security<br>• **Judgement:** a diverse mix matched to uses (electricity vs heat vs transport) with storage is most sustainable<br>– transition must start before depletion<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• Any accurate, relevant material that is not in the indicative content. | 9 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | • **sustainability** → meeting present needs without stopping future generations meeting theirs<br>• **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous<br>• **base load** → the steady minimum demand that must always be met<br>**Do not credit**<br>• carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen<br>• 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do | |
| | **Model answer - Level 3 (8 marks)** | |
| | Whether a resource suits a use depends on its features: renewable or not, abundance, energy density, locational constraints, intermittency and predictability, the conversions needed, ease of storage and transport, how developed the technology is and its impacts. Fossil fuels dominate because they combine high energy density, easy storage and transport, high temperatures for industry and mature infrastructure - which is why petrol and diesel still monopolise transport. But they are finite, cause acid rain, smoke and $CO_2$, and are politically concentrated. Nuclear fuel has the highest energy density and gives reliable base load with few deliveries, yet the technology is complex and expensive and produces long-lived waste against public opposition. | |
| | Renewables have the opposite profile: low energy density needing large collectors, intermittency (wind and solar unpredictable, tidal predictable), strong locational constraints and no direct storage, so they need conversion, storage and grid links and cannot yet reach fossil-fuel temperatures or replace liquid fuels. Sustainability must count impacts before use - extraction, processing, embodied energy of equipment - as well as pollution, habitat damage and depletion during use, and economics and politics (subsidies, planning, external costs left out of prices, energy security) shape what is actually chosen. | |
| | No single resource is suitable for everything. Electricity, heat and transport have different requirements, so the most sustainable system is a diverse mix matched to each use with storage, and the transition must begin before depletion forces it. | |
| | **Weaker answer - Level 1 (2 marks)** | |
| | Energy resources have different features. Fossil fuels are non-renewable and will run out but they have a high energy density. Renewable energy like wind and solar will not run out but they are intermittent because the wind does not always blow. Nuclear power produces radioactive waste. Renewable energy is more sustainable because it does not produce carbon dioxide. So we should use more renewable energy. | |
| | **What makes the difference** | |
| | • Level 1 mentions two or three features; Level 3 works through the full set and links each to a specific use (transport, base load, industry). | |
| | • Level 3 includes sustainability concepts (impacts before and during use, external costs) and political factors that Level 1 omits entirely. | |
| | • The conclusion in Level 3 is a reasoned position (a diverse mix matched to uses); Level 1's 'use more renewables' does not follow from a discussion. | |

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| Question | Marking guidance | Marks |
|---|---|---|
| **10.2** | 1 mark for the point below.<br>**1** Very high energy density fuel means only a few tens of tonnes are needed each year, compared with 10 000 t of coal per day for a coal-fired station<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-182; AQA spec 3.3.2-3.3.3. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| 11.1 | **Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• **Composition and structure:** $N_2$, $O_2$, Ar, $CO_2$, ozone<br>– troposphere and stratosphere<br>– pressure and liquid water<br>– gases for biological molecules and industry<br>• **Energy processes:** UV absorbed by O / $O_2$ / $O_3$ dynamic equilibrium<br>– visible light absorbed at the surface<br>– infrared absorbed by greenhouse gases (33 °C warming)<br>– heat distribution by winds and ocean currents<br>– transport of water vapour<br>– energy budget and albedo<br>• **Global climate change:** anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species<br>– difficulties of monitoring and prediction<br>– control strategies and agreements<br>• **Ozone depletion:** CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time<br>• **Other threats:** aerosols, smoke, land-use change<br>– interconnection of processes | 25 |
| | • **Evaluation:** which functions are most at risk<br>– success of ozone controls vs slow progress on greenhouse gases<br>– justified conclusion<br>*Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*<br>**Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| 11.2 | **Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• **Context:** uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links<br>• **New sources:** rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers<br>• **Management:** artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation<br>• **Conservation:** low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering<br>– pollution control to protect sources<br>• **Treatment:** matching quality to use<br>– screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis<br>• Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion<br>*Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* | 25 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5. | |

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